



LIBRARY MANAGEMENT SYSTEM DEVELOPED USING PYTHON OOP

Optimizing Your Library Operations

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SUBJECT: PYTHON PROGRAMMING

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Introduction:

The Library Management System is a project developed in Python, leveraging Object-Oriented Programming (OOP) principles. This system helps users with various tasks, including:

- Managing books
- Borrowing books
- Returning books
- Searching for books
- Registering users

This project highlights the practical use of Python programming concepts.

Project Objectives And Tools – technologies used :

- Understand the fundamentals of Object–Oriented Programming (OOP)
- Develop a basic library management system
- Manage books and user accounts efficiently
- Integrate features for borrowing and returning items
- Improve Python programming skills

Tools and Technologies Used :

- Python :– Programming Language
- OOP :– Program Structure
- VS Code / PyCharm :– Code Editor
- Terminal :–Program Execution

OOP Concepts Utilized:

Concepts Used:

- Class
- Object
- Constructor
- Methods
- Encapsulation
- Association

Classes Used:

- Book
- User
- Library

Library Management System Classes :

Book Class:

Attributes: book_id, title, author, quantity

Methods: display_book_info(), check_availability(), update_quantity()

Purpose: Stores and manages book information.

User Class:

Attributes: user_id, name, borrowed_books

Methods: borrow_book(), return_book(), view_borrowed_books()

Purpose: Handles user activities and borrowed books.

Library Class:

Attributes: books, users

Methods: add_book(), register_user(), search_book_by_title(), list_books()

Purpose: Controls the overall library management system.

System Workflow:

- Create Library
↓
- Add Books
↓
- Register Users
↓
- Display Menu
↓
- Borrow / Return Books
↓
- Update Quantity
↓
- Exit System

Main Features:

- View– all available books
- Add– new titles to the collection
- Search– for books by their title
- Borrow– books for your use
- Return– borrowed books
- View– a list of borrowed books
- Register– new users
- Prevent– the creation of duplicate IDs

Sample Code and output :

Borrow Book Function

```
```python
if book.check_availability():
 self.borrowed_books.append(book)
 book.update_quantity(-1)
```
```

Explanation:

- Checks availability
- Adds book to the borrowed list
- Reduces quantity

Program Output

Welcome to the Library Management System

- 1.View all books
- 2.Add books
- 3.Search for a book by title
- 4.Borrow a book
- 5.Return a book
- 6.View borrowed books
- 7.Add new User
- 8.Exit

Advantages and Limitations :

Benefits of the Project

- User-friendly interface
- Well-organized structure
- Streamlined book management
- Practical implementation of OOP principles
- Suitable for beginners

Constraints

- Lacks a database connection
- No graphical user interface (GUI)
- Data is not stored permanently
- Absence of a login system

Future Enhancements:

- Incorporate database support
- Implement a graphical user interface (GUI)
- Establish a login system
- Introduce a due date management system
- Integrate fine calculation
- Enable file handling

Conclusion:

The Library Management System effectively showcases the practical application of Python and Object-Oriented Programming (OOP) concepts. This project has enhanced understanding of:

- Classes and Objects
- Functions
- Lists
- Loops
- Real-world system development

THANK YOU FOR
LISTENING! ✦